Clinical–genomic residual prediction of lung function

Code-verified UKB architecture | common genomic encoder, population-specific fusion and training

A Shared genomic representation (instantiated separately in the two population models)

Genotype dosage matrix

4,064 fixed variants
Allele alignment; mean imputation
Training-local standardization

Genomic patch tokens

Within-chromosome patches
64 variants → 16-dimensional token
+ chromosome + position encoding

Transformer encoder

1 encoder layer; 2 attention heads
Embedding 16; feed-forward 32
Dropout 0.35; pre-layer norm

Task-query attention pooling

Encoded SNP tokens + 4 PRS tokens
Layer norm → learned query weights
Task-specific 16-dimensional h_j

PRS are derived from the same dosages.
They join after SNP self-attention,
not before the Transformer encoder.

Four PRS inputs → scalar projection + PRS-type embedding

GWAS-weighted dosage scores at $P \le 10^{-5}$, 10^{-4}, 10^{-3} and 0.01

B Healthy model | six-task joint neural correction

clinical baseline bypass b_j

16 clinical inputs

Spirometry, demographics, smoking,
blood counts and prediction interval
Ridge (continuous) / logistic (risk)
→ six clinical baseline predictions

Clinical + genomic neural fusion

Clinical MLP: n → 32 → 16
Concatenate clinical vector with h_j
Six task heads: 32 → 16 → 1
Residual heads initialized at zero

Residual correction

Continuous: $b_j + w_j\, s_j\, \delta_j$
Risk: sigmoid(logit(b_j) + $w_j\, \delta_j$)
Validation selects checkpoint
and task-specific residual weight

Six outputs

Next FEV_1, FVC, FEV_1/FVC
+ three rapid-decline risks
Independent classification heads
Fixed test: n = 1,899

Joint loss: masked mean of 3 standardized residual MSE terms + 3 BCE-with-logit-offset terms.

C COPD model | clinical + 31 SNP baseline, separate neural regression / classification fits

clinical baseline bypass b_j

Clinical + 31 lead SNP inputs

Task-specific clinical feature sets
Affine Ridge / logistic baseline
Three-fold inner OOF predictions
for neural residual targets

Genomic residual networks

Same genomic encoder as A
Regression: 3 task queries / heads
Classification: separate 3 heads
No clinical MLP in this branch

Two-seed mean + fusion

Continuous: $b_j + \mu_j + s_j\, \delta_j$
Risk: sigmoid(logit(b_j) + δ_j)
60 epochs; fixed residual weight 1
No outer-case model selection

Six outputs

Same three continuous targets
+ three rapid-decline risks
Leave-one-participant-out CV
UKB COPD: n = 2,067

Separate fits: 3-task residual MSE; 3-task BCE with clinical logit offsets and entropy normalization.

b: clinical baseline; δ: neural correction; s / μ: training residual scale / mean; w: validation-selected weight (Healthy). OOF = out-of-fold.

Healthy / COPD are cohort strata, not classification labels. Rapid decline is defined from annualized observed change. Historical variant discovery overlap remains.